

Testcase ID	Testcase (Logic Branch Targeted)	Input parameters	Expected outcome	Actual outcome	Result
Test Case ID	Method/Branch Path	Input/Scenario	Expected Outcome	Actual Outcome	Status
A.01.01	validateCredentials()(id == null)	null studentID, any otp	Exception: "Null Credentials Structure"	System threw NullCred	PASS
A.01.02	validateCredentials()(otp == null)	Valid studentID, null otp	Exception: "Null Credentials Structure"	System threw NullCred	PASS
A.01.03	validateCredentials()(Success)	Valid studentID, Matching otp	Return Boolean true	Method returned true;	PASS
A.01.04	validateCredentials()(No ID)	studentID = "EX999"	Exception: "Student Record Not Found"	System threw Student	PASS
A.01.05	validateCredentials()(Wrong OTP)	Valid studentID, otp = "0000"	Return Boolean false	Method returned false	PASS
A.01.06	validateCredentials()(Locked)	Valid studentID, wrong otp, retries=3	Exception: "Account Locked"	Account status update	PASS
A.01.07	saveAnswer() (Valid MCQ)	qType=MCQ, answer="1"	Acknowledge Save	Response updated in I	PASS
A.01.08	saveAnswer() (MCQ Bound)	qType=MCQ, answer="9"	Exception: "Invalid MCQ Index Selected"	Threw InvalidMCQInde	PASS
A.01.09	saveAnswer() (Valid Text)	qType=TEXT, "A short answer."	Acknowledge Save	Text content successfi	PASS
A.01.10	saveAnswer() (Text Bound)	answer.length() > 5000	Exception: "Answer Text Length Exceeds..."	Threw AnswerTextLen	PASS
A.01.11	saveAnswer() (DB Crash)	Valid inputs, DB Connection Lost	Exception: "Response Persistence Failed"	Caught SQLException	PASS
A.01.12	finalizeSubmission()(Locked)	Valid inputs, examStatus == "Locked"	Exception: "Session Locked - Cannot Finalize"	Prevented finalize() ca	PASS
A.01.13	finalizeSubmission()(Manual)	Valid inputs, examStatus == "Active"	Exam status updated to "Submitted"	Status transitioned to	PASS
A.01.14	getQuestions() (No Exam)	examID not found in database	Exception: "Exam ID Does Not Exist"	Threw ExamIDDoesNo	PASS
A.01.15	State Mismatch (Race Condition)	Simultaneous 'save' during update	Exception: "Operational Failure (Race Condition)"	Concurrency control d	FAIL